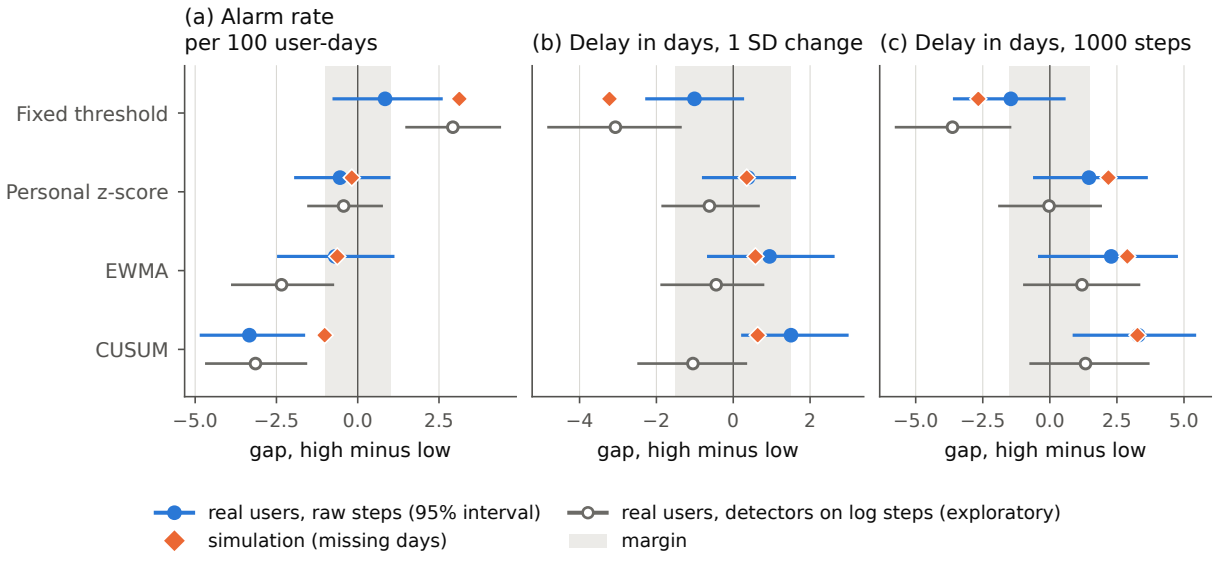
(a) Alarm rate
per 100 user-days
(b) Delay in days, 1 SD change
(c) Delay in days, 1000 steps
Fixed threshold
Personal z-score
EWMA
CUSUM
−5.0
−2.5
0.0
2.5
−4
−2
0
2
−5.0
−2.5
0.0
2.5
5.0
gap, high minus low
gap, high minus low
gap, high minus low
real users, raw steps (95% interval)
real users, detectors on log steps (exploratory)
simulation (missing days)
margin